

A Reflection of NDNComm 2020 NDN 10-Year Lessons Learned Panel

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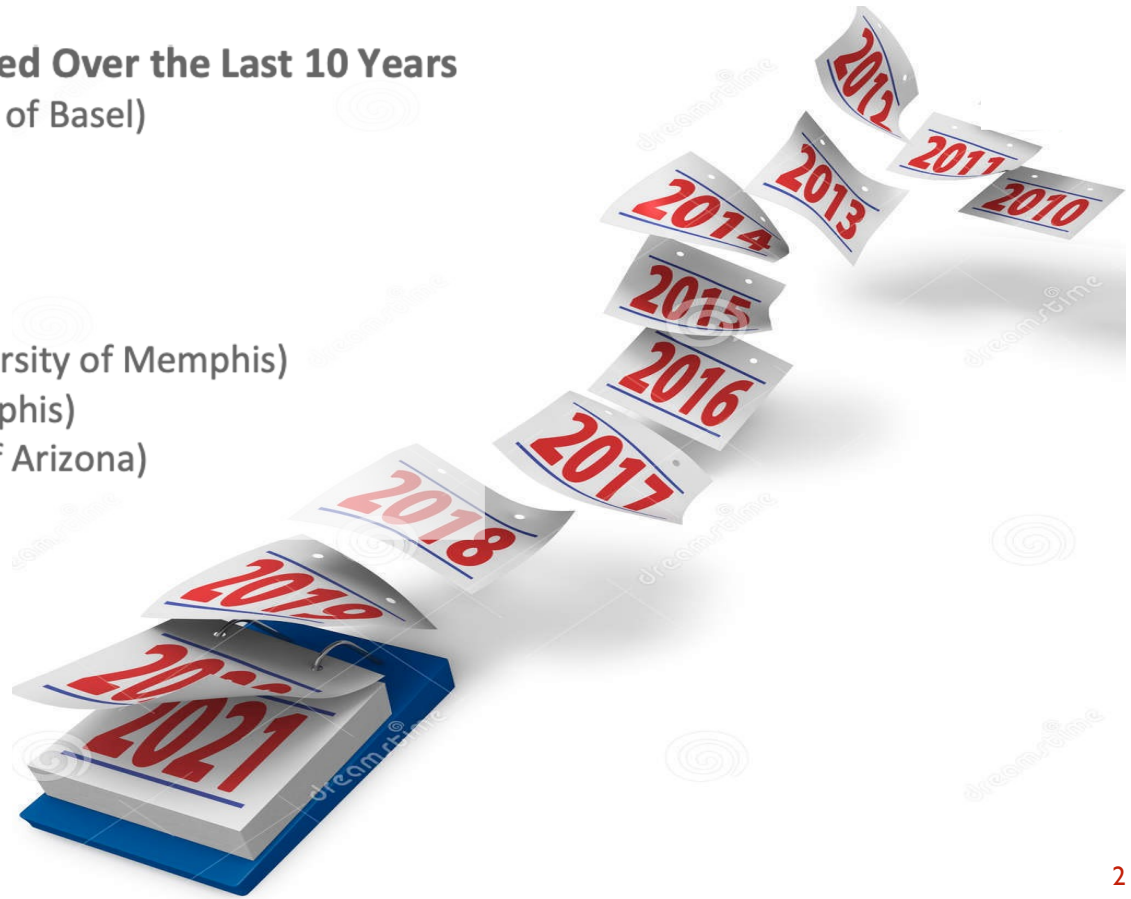
UCLA

NDN COMMUNITY MEETING 2021

Years flying by when we are having fun

Panel 3: Named Data Networking: Lessons Learned Over the Last 10 Years

Moderator: Christian Tschudin (University of Basel)
Panelists: Jeff Burke (UCLA REMAP)
David Clark (MIT)
Darleen Fisher (NSF)
Van Jacobson (Google)
Christos Papadopoulos (University of Memphis)
Lan Wang (University of Memphis)
Beichuan Zhang (University of Arizona)
Lixia Zhang (UCLA)





DNcomm Panel 3

**NAMED DATA
NETWORKING**



Round 1

What are highlights of the last ten years?

Round 2

Where things could be improved

Round 3, closing statements

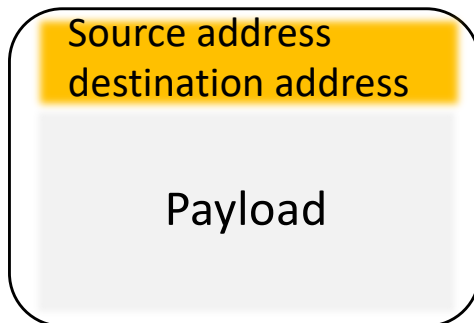
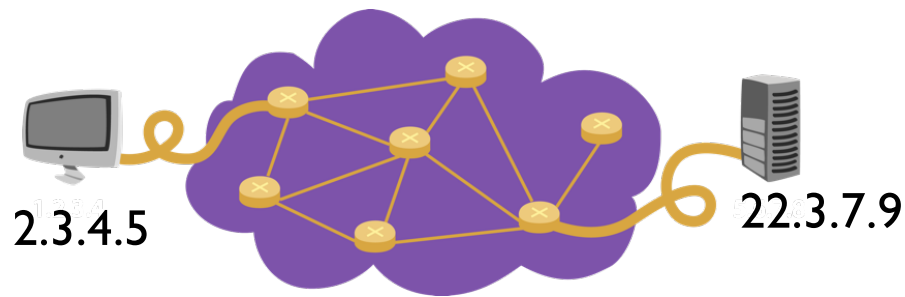
Where are we now?



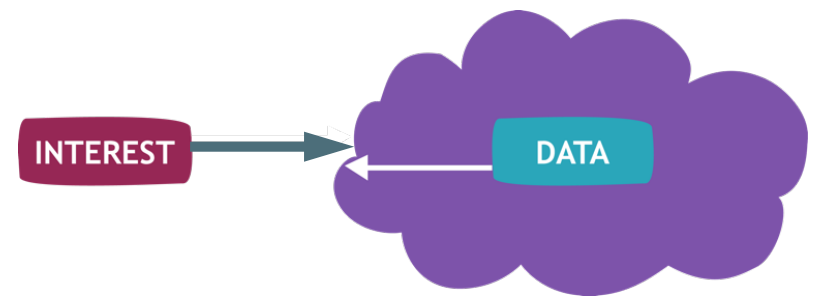
<https://www.youtube.com/watch?v=oCZMoY3q2uM>

Highlight #1

A bold, simple new way to networking



IP packet



Application data name
(a few optional parameters)

Interest packet

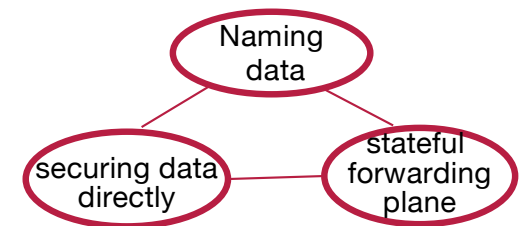
Application data name
(a few optional metadata)
data
crypto signature

Data packet

Highlight #2

Semantically named, secured data

- NDN got its basic architectural idea right: *semantically named, secured data (grams)* as the basic architectural building block
- Networking: delivering bags of bits
 - Movies, sport streaming, games
 - IoT sensing data, command to lock the door
 - Crypto keys, certificates, security policies
- Semantic names reflect application context, relations among networked entities → lay the ground for context-specific security solutions
 - The development of trust schema



Applicable, has been applied to solving problems in broad areas

- Internet of Things,
- Vehicular networking
- Distributed social apps
- Disruption-tolerant networking
- Satellite networking
- Edge computing

Highlight #3

Commitment from NSF & community

- NDN was funded under the NSF Future Internet Architecture (FIA) program in 2010
 - Continued support and encouragement from NSF over the decade
- “A Dedicated Persistent Team”
- Commitment by the broader community

Lessons learned #1

Where to start a new architecture

- The 10-year journey started with the FIA charging order :
developing a new Internet architecture
 - Supporting Internet at scale
 - Naming data and stateful forwarding are such novel concepts, got bombarded with questions “how can they possibly scale”
- Designing a new architecture requires seeing the overall picture
 - Solely focusing on network forwarding layer would not lead to a departure from IP
- A new architecture starts from the edge, not the center
 - 2017 PI meeting set future direction: edge in

Lessons learned #2

Understanding our own limitations

- Van: “The research is done in the researchers’ comfort zone”
- The layered protocol architecture seems segregated not only the functional modules, but also people’s expertise
 - Today’s education train specialists, most NDN PIs with network background
- Developing a new architecture offers the best training for generalists
- Reaching out to people with real problems to solve

Lessons learned #3

How to develop a new architecture

- Ultimate goal of networking: enabling applications
- Application-driven architecture development
 - Solving real problems forces architectural details to be filled in and, most importantly, verifies and shapes the architectural direction
- Building usable apps over NDN, attracting app developers to NDN
 - High level API
 - Usable security
 - Plug-n-play
- Growing the community

Where we are now

- It takes time to roll out a new architecture
 - IP: 1974 Cerf-Kahn paper → 1981 TCP/IP spec → early 90's web rollout
- NDN rollout: at the end of beginning
 - Learned where the difficulties are in pushing out NDN
 - Learned how to get there: edge-in
 - “edge” along 2 dimensions: network, applications
- Using apps to drive the architecture development and rollout
 - Does the architecture do the right thing
 - What's missing

This is a community meeting

- Sharing achievements over the year
- Identifying new issues
- Collecting inputs to formulate the marching order for next year